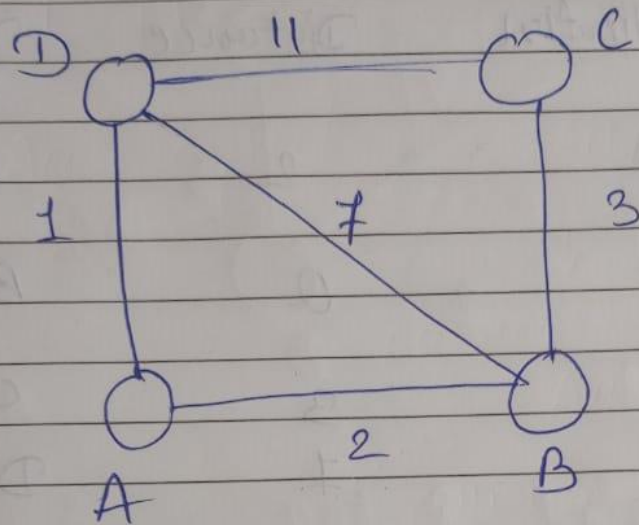


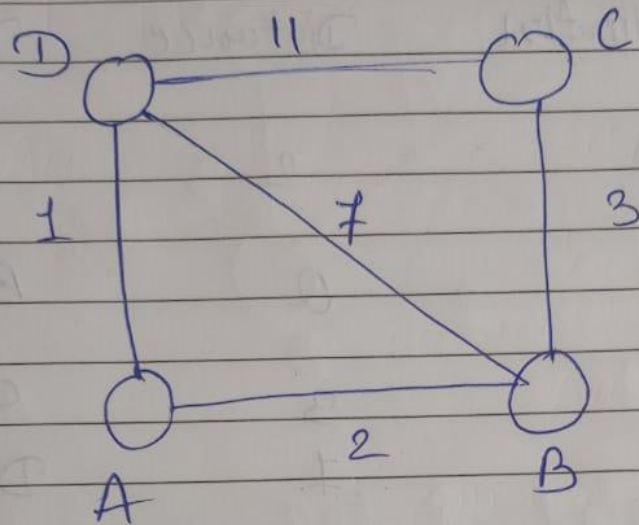
Distance Vector Algorithm



A# A

Dest ⁿ	Distance	Next Hop
A	0	A
B	2	B
C	∞	—
D	1	D

Distance Vector Algorithm



A# A

Dest ⁿ	Distance	Next Hop
A	0	A
B	2	B
C	∞	—
D	1	D

At Router B

Destination	Distance	Next hop
A	2	A
B	0	B
C	3	C
D	1	D

At Router C.

Dest ⁿ	Distance	Next hop
A	∞	—
B	3	B
C	0	C
D	1	D

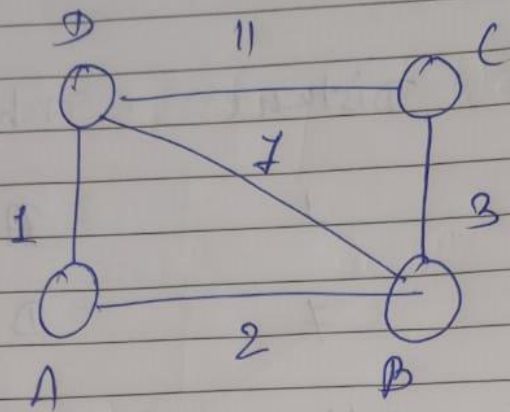
At Router D

Dest ⁿ	Distance	N ^o hope
A	1	A
B	7	D
C	11	C
D	0	D

After collecting the neighbours

for A.

<u>B</u>	<u>D</u>	<u>Destⁿ</u>	<u>Distⁿ</u>	<u>N^o H</u>
2	1	A	0	A
0	7	B		
3	11	C		
7	0	D		



In Next Routing.
 for Routing A
A → B

option-1 {

$$A \rightarrow D \quad D \rightarrow B$$

$$1 \quad + \quad 7 \quad = \quad 8$$

Min
value →

option-2 {

$$A \rightarrow B \quad B \rightarrow B$$

$$2 \quad + \quad 0 \quad = \quad 2$$

4

Next Round

Next A to C

$A \rightarrow C$

$$\text{option-1} \left\{ \begin{array}{l} A \rightarrow D \quad D \rightarrow C \\ 1 \quad + \quad 11 = 12 \end{array} \right.$$

Min
value

$$\text{option-2} \left\{ \begin{array}{l} A \rightarrow B + B \rightarrow C \\ 2 \quad + \quad 3 = 5 \end{array} \right.$$

Next A to D

$$\left\{ \begin{array}{l} A \rightarrow D \quad D \rightarrow D \\ 1 \quad + \quad 0 = 1 \end{array} \right.$$

$$\text{Min value} \left\{ \begin{array}{l} A \rightarrow B \quad B \rightarrow D \\ 2 \quad + \quad 7 = 9 \end{array} \right.$$

so After completion of Routing.

final Routing Table.

<u>Destⁿ</u>	<u>Distⁿ</u>	<u>Next hope</u>
A	0	A
B	2	B
C	5	B
D	1	D

